

Prerequisites

Prop. Are abelian categories Ab , Mod_A , $\text{Sh}(X)$ (gorups), $\text{QCoh}(X)$ and $\text{Coh}(X)$.

Def (Delta functor). $F : \mathcal{C} \rightarrow \mathcal{D}$ left-exact. Exist F^i and $\delta^i : F^i(C) \rightarrow F^{i+1}(A)$ for all sas $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ such that

$$\cdots \rightarrow F^i(A) \rightarrow F^i(B) \rightarrow F^i(C) \xrightarrow{\delta^i} F^{i+1}(A) \rightarrow \cdots$$

is exact. It is universal if it is initial in the category of delta functors with $F^0 = F$.

Def (Erasable). δ is erasable if for all A there exists a morphism u such that $F^i(u) = 0$. Every universal delta functor is erasable.

Def. $\mathcal{C}$ Abelian.

- **Injective** $I \in \mathcal{C}$ if $\text{Hom}(-, I)$ is exact.
- **Projective** $P \in \mathcal{C}$ if $\text{Hom}(P, -)$ is exact.
- **Enough injectives** if for all A there exists an injection $A \rightarrow I$ with I injective.
- **Enough projectives** if for all A there exists a surjection $P \rightarrow A$ with P projective.

Prop. The categories Ab , Mod_A , $\text{Sh}(X)$ have enough injectives. $\text{QCoh}(X)$ and $\text{Coh}(X)$ do not.

Prop. I injective, then $0 \rightarrow I \rightarrow A \rightarrow B \rightarrow 0$ splits.

Prop. We only need to check $F(I) = 0$ for injectives to check erasability.

(Commutative lemmas)

Def. Right derived functor $R^i F(A) := H^i(F(I^\bullet))$ where $A \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots$. These are erasable.

Def (Sheaf cohomology). $H^i(X, -) := R^i \Gamma(X, -)$.

Def. F left-exact, $\mathcal{C}$ enough injectives. $A \in \mathcal{C}$ acyclic if $R^i F(A) = 0$ for $i \leq 1$.

Def (Flasque). Restriction morphisms are surjective. Flasque sheaves are acyclic.

Prop. $\mathcal{F}$ injective $\Rightarrow \mathcal{F}$ flasque.

Def (Cech). $\mathcal{F}$ presheaf, $\mathcal{U} = \{U_i\}$ open cover of X . $C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \cdots < i_p} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p})$.

$$0 \rightarrow C^0(\mathcal{U}, \mathcal{F}) \rightarrow C^1(\mathcal{U}, \mathcal{F}) \rightarrow \cdots, \quad (ds)_{j_0, \dots, j_{p+1}} = \sum_{k=0}^{p+1} (-1)^k s_{j_0, \dots, \hat{j}_k, \dots, j_{p+1}}|_{U_{j_0, \dots, j_{n+1}}}$$

Define $\check{H}^i(\mathcal{U}, \mathcal{F}) = H^i(C^\bullet(\mathcal{U}, \mathcal{F}))$.

Theorem. $\mathcal{F}$ sheaf and $\mathcal{U}$ affine cover of X , then $\check{H}^i(\mathcal{U}, \mathcal{F}) = H^i(X, \mathcal{F})$.

Theorem (Serre's criterion). X qc, then TFAE:

- X affine
- $H^i(X, \mathcal{F}) = 0$ for all $i > 0$ and $\mathcal{F} \in \text{QCoh}(X)$

- $H^1(X, \mathcal{F}) = 0$ for all $\mathcal{F} \in \text{QCoh}(X)$

Prop. X projective of dimension n over k , then $H^i(X, \mathcal{F}) = 0$ for all $i > n$ and $\mathcal{F} \in \text{QCoh}(X)$.

Theorem (Cohomology of projective).

$$H^q(\mathbb{P}_A^n, \mathcal{O}(d)) = \begin{cases} A[x_0, \dots, x_n]_d & q = 0 \\ 0 & q \neq 0, n \\ (\frac{1}{x_0 \cdots x_n} A[x_0^{-1}, \dots, x_n^{-1}])_d & q = n \end{cases}$$

Theorem. A Noetherian, $\mathcal{F}$ coherent on $X = \mathbb{P}_A^n$ (or projective over A). Then

- $H^i(X, \mathcal{F})$ is f.g. A -module for $i \geq 0$.
- $H^i(X, \mathcal{F}(d)) = 0$ for $d \gg 0$ and $i > 0$.

Prop. Let $f : X \rightarrow Y$ and $\mathcal{F} \in \text{Sh}(X)$. Then, $R^i f_*(\mathcal{F})$ is the sheaf associated to the presheaf $V \mapsto H^i(f^{-1}(V), \mathcal{F})$. Also

$$(R^i f_* \mathcal{F})|_V \cong R^i (f_V)_*(\mathcal{F}|_{f^{-1}(V)})$$

Prop. Flasque sheaves are f_* -acyclic.

Prop. $f : X \rightarrow Y$ affine, $\mathcal{F} \in \text{QCoh}(X)$. Then, $H^i(X, \mathcal{F}) \cong H^i(Y, f_* \mathcal{F})$.

Prop. $f : X \rightarrow Y$ qcqs, $\mathcal{F} \in \text{QCoh}(X)$. Then,

- $R^i f_* \mathcal{F} \in \text{QCoh}(Y)$ for all $i \geq 0$.
- If Y affine, $R^i f_* \mathcal{F} \cong H^i(X, \mathcal{F})^\sim$

Prop. $f : X \rightarrow Y$ projective between Noetherians. $\mathcal{F} \in \text{Coh}(X)$, then $R^i f_* \mathcal{F} \in \text{Coh}(Y)$.

Def (Homs). $\mathcal{F} \in \text{Mod}(X)$.

- $\text{Ext}_{\mathcal{O}_X}^i(\mathcal{F}, -)$ right derived functor of $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, -) : \text{Mod}(X) \rightarrow \text{Ab}$.
- $\mathcal{E}xt_{\mathcal{O}_X}^i(\mathcal{F}, -)$ right derived functor of $\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, -) : \text{Mod}(X) \rightarrow \text{Mod}(X)$.

Prop. $\mathcal{G} \in \text{Mod}(X)$

- $\text{Ext}^i(\mathcal{O}_X, \mathcal{G}) \cong H^i(X, \mathcal{G})$
- $\mathcal{E}xt^i(\mathcal{O}_X, \mathcal{G}) \cong 0$ for $i > 0$ and $\cong \mathcal{G}$ for $i = 0$.

Prop. Given ses $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ in $\text{Mod}(X)$, there is long exact sequence

$$0 \rightarrow \text{Hom}(\mathcal{F}'', \mathcal{G}) \rightarrow \text{Hom}(\mathcal{F}, \mathcal{G}) \rightarrow \text{Hom}(\mathcal{F}', \mathcal{G}) \rightarrow \text{Ext}^1(\mathcal{F}'', \mathcal{G}) \rightarrow \cdots$$

Prop. $\mathcal{F} \in \text{Mod}(X)$ and $\cdots \rightarrow \mathcal{L}_1 \rightarrow \mathcal{L}_0 \rightarrow \mathcal{F} \rightarrow 0$ loc. free of finite rank. Then

$$\mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \cong H^i(\text{Hom}(\mathcal{L}_\bullet, \mathcal{G}))$$

Prop. $\mathcal{L}, \mathcal{F}, \mathcal{G} \in \text{Mod}(X)$ and $\mathcal{L}$ loc. free of finite rank. Then,

- $\mathcal{E}xt^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \otimes \mathcal{L}^\vee \cong \mathcal{E}xt^i(\mathcal{L}, \mathcal{G} \otimes \mathcal{L}^\vee)$
- $\text{Ext}^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \text{Ext}^i(\mathcal{F}, \mathcal{G}) \otimes \mathcal{L}^\vee$

Prop. $x \in X$ a point in a nbh U with a presentation $\mathcal{O}_U^{\oplus m} \rightarrow \mathcal{O}_U^{\oplus n} \rightarrow \mathcal{F}|_U \rightarrow 0$. Then

$$\text{Hom}(\mathcal{F}, \mathcal{G})_x \cong \text{Hom}_{\text{Mod}(\mathcal{O}_{X,x})}(\mathcal{F}_x, \mathcal{G}_x)$$

Prop. $x \in X$ a point in a nbh U with a resolution $\cdots \rightarrow \mathcal{L}_1 \rightarrow \mathcal{L}_0 \rightarrow \mathcal{F}|_U \rightarrow 0$ with $\mathcal{L}_i$ loc. free of finite rank. Then

$$\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})_x \cong \text{Ext}_{\text{Mod}(\mathcal{O}_{X,x})}^i(\mathcal{F}_x, \mathcal{G}_x)$$